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RESEARCH ARTICLE

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Air–Sea CO_2 Fluxes at Coastal Hydrothermal Vents in Deception Island (Antarctica)

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Key Points:

- Monitored nearshore hydrothermal sites show predominantly positive air–sea CO_2 fluxes and intense outgassing pulses at tidal timescales
- Local changes in p_{CO_2} in seawater are mainly driven by tidal and wind-waves mixing, wind-induced advection, and meltwater runoff
- Exceptionally high air–sea flux pulses, which typically occur at tidal timescales, range between ∼130 mmol m^{-2} d^{-1} and 11, 260 mmol m^{-2} d^{-1}

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract This study investigates the spatio-temporal variability of seawater partial pressure p_{CO_2} and air–sea CO_2 fluxes at four coastal hydrothermal sites (viz. Pendulum Cove, Fumarole Bay, Whalers Bay, and Murature Beach) in Port Foster (Deception Island, Antarctica) during the 2024 and 2025 austral summers. Time-series at all sites exhibited p_{CO_2} background levels that exceeded atmospheric concentrations (∼418 ppm in 2024 and ∼421 ppm in 2025). Pendulum Cove, Fumarole Bay, and Whalers Bay showed remarkably similar background p_{CO_2} levels (600 − 640 μatm). Murature Beach showed lower levels (∼300 μatm), suggesting enhanced biological uptake from phytoplankton biomass. Recurrent p_{CO_2} pulses over background levels were observed. Murature exhibited the highest pulses of both p_{CO_2}, exceeding 80,000 μatm, and air–sea CO_2 fluxes, reaching ∼11,260 mmol m^{-2} d^{-1}. These fluxes were orders of magnitude higher than those at the other sites (∼134 − 529 mmol m^{-2} d^{-1}). They also substantially exceeded fluxes reported for other polar and coastal regions and were comparable to those observed in other volcanic systems. Air–sea CO_2 fluxes in Port Foster originate from biological CO_2 uptake and hydrothermal inputs, which can temporarily overwhelm biological uptake. Tides, wave-driven mixing, wind-induced advection, and meltwater runoff are the main physical drivers of p_{CO_2} variability at coastal hydrothermal vents. Tides modulate these pulses: higher p_{CO_2} levels and air–sea fluxes are triggered at low tide, whereas high tide brings concentrations and fluxes back toward local background levels. Overall, p_{CO_2} dynamics and carbon exchange in polar hydrothermal environments reflect a site-dependent balance among hydrothermal supply, physical forcing, and biological activity.

Plain Language Summary This study quantifies natural CO_2 emissions from coastal hydrothermal sites at Port Foster (Deception Island, Antarctica), important for CO_2 global stocks. Measurements were recorded at four sites (Pendulum Cove, Fumarole Bay, Whalers Bay, and Murature Beach) during the austral summers of 2024 and 2025. Results show that CO_2 concentrations are strongly influenced by local physical processes, such as tides, waves, wind-driven currents, and glacial meltwater. Mixing driven by tides, waves, and wind generally reduced CO_2 levels by diluting hydrothermal inputs. At all study sites, CO_2 concentrations in surface water regularly exceeded those in the atmosphere. Murature Beach showed particularly extreme variability, combining the lowest background CO_2 levels with the highest CO_2 peaks and air–sea fluxes. The lowest background level was partly driven by phytoplankton CO_2 uptake, whereas the extreme peaks likely resulted from inputs of CO_2-rich hydrothermal waters that temporarily overwhelmed this biological uptake. These findings reveal that CO_2 fluxes in these coastal Antarctic environments are governed by a balance between hydrothermal inputs, local physical forcing and biological activity.

1. Introduction

Carbon cycling in the Southern Ocean (SO) is fundamental to Earth's climate. By absorbing atmospheric carbon dioxide (CO_2) through biological and solubility pump processes and releasing deep ocean CO_2 (Gruber et al., 2019), the SO modulates Earth's climate on seasonal-to-millennial timescales. The SO acts thus as a major sink of excess heat and anthropogenic carbon, mitigating global warming effects (Frölicher et al., 2015; Terhaar et al., 2021). However, this mitigation has ecosystem costs. Biogeochemical cycling is directly affected by the decline in seawater pH by CO_2 absorption, that is, ocean acidification, and indirectly via the heat effect on sea-ice dynamics, glacial meltwater inputs, winds and ocean circulation (Henley et al., 2017; Kerr et al., 2018).

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As biogeochemical processes vary strongly, particularly between Antarctic coastal/shelf areas and the open ocean, limited sampling fails to capture their high spatiotemporal variability in the SO (Fay et al., 2024). Hence, accurate local observations of biogeochemical tracers and processes are urgently needed to clarify their role in future climate and ecosystem trends. However, logistical challenges and extreme weather limit observational coverage (Bakker et al., 2016). Filling observation gaps in Antarctic shelf waters is critical, as they are shaped by both continental and offshore mechanisms, whether natural and/or anthropogenic.

The Western Antarctic Peninsula (WAP) is among the most rapidly changing shelf subregions due to climate change. This subregion is experiencing accelerated sea-ice loss compared to other Antarctic regions (Stammerjohn & Scambos, 2020), reducing surface ocean CO_2 uptake through reduced stratification (Droste et al., 2025). Moreover, increasing anthropogenic CO_2 emissions contribute to global warming, cryosphere loss, ocean acidification, and alterations in the marine carbon cycle (Morley et al., 2020). Within the South Shetland Islands archipelago along the WAP lies Deception Island, an active volcano with a semi-enclosed basin known as Port Foster. As one of the most active volcanoes on the Antarctic continent, with numerous eruptive events recorded over recent centuries (Geyer et al., 2023), Deception Island is, in fact, the only maritime volcano among Antarctica's four geothermally active systems featuring surface hydrothermal activity (Herbold et al., 2014). Deception Island is a unique yet sensitive ecosystem (Angulo-Preckler et al., 2020; Barbosa et al., 2012; Duarte et al., 2021), with fumaroles and hydrothermal vents that release substantial amounts of greenhouse gases, including CO_2 and methane (CH_4) (Bartolini et al., 2014; Hopfenblatt et al., 2021; Ibáñez et al., 2003).

Biogeochemical studies at Deception Island have focused mainly on land. Time series of partial pressure of CO_2 of seawater (p_{CO_2}) and associated air–sea fluxes at coastal fumarole sites have not been computed to date. Detailed physical and biogeochemical characterization of nearshore water masses remains lacking, and no consistent multi-site time-series studies exist on Deception Island to link atmospheric and oceanographic forcing with CO_2 observations. While hydrothermal activity is magmatic in origin, the dispersion of emitted gases in water masses is primarily driven by hydrodynamics (Caselli et al., 2002). Valentin et al. (1989) reported that CO_2 dominates hydrothermal gases on Deception Island (75–90%). Although previous basin-scale observations indicate that Port Foster, as a whole, can act as a net surface CO_2 sink, shallow waters directly influenced by hydrothermal vents may behave as intense local CO_2 sources (Belyaev et al., 2024).

This study aims to provide a coherent view of the spatio-temporal variability of seawater p_{CO_2} and associated air–sea CO_2 fluxes in shallow hydrothermal areas of Port Foster. Studying fumarole-influenced sites in Deception Island provides a natural laboratory to investigate how geothermal CO_2 emissions interact with oceanographic forcings. These underrepresented environments in global carbon budgets may exert disproportionately high local fluxes with regional implications. Specifically, the objectives are to: (a) unravel joint variability between the spatial and temporal p_{CO_2} patterns and the hydrographic and biogeochemical conditions of the water column; (b) quantify the magnitude and variability of local air–sea CO_2 fluxes; and (c) assess the relative roles of physical forcings acting at different timescales, such as tides, wind, waves and meltwater inputs and local biogeochemical processes in controlling the observed p_{CO_2} and air–sea flux variability.

To address the objectives, high frequency records of seawater p_{CO_2}, temperature, salinity, dissolved oxygen, sea level, wind, and wave height were collected in Port Foster during the austral summers of 2024 and 2025. In addition, discrete seawater samples were collected to determine chlorophyll-*a*, pH and total alkalinity, providing complementary biogeochemical information for interpreting the p_{CO_2} records. Air–sea fluxes were determined using p_{CO_2} and wind data.

This study is organized as follows: Section 2 describes the study area and measurements. Section 3 presents the spatio-temporal variability of biogeochemical properties and air–sea CO_2 fluxes estimates. Section 4 discusses these results and compares estimated fluxes with other regions, and Section 5 summarizes the main conclusions.

2. Material and Methods

2.1. Study Area

Deception Island, located in the South Shetland Islands north of the Antarctic Peninsula (Figure 1), is a horseshoe-shaped stratovolcano and one of Antarctica's most active. Its flooded caldera, Port Foster, forms a semi-enclosed basin of ~30 km perimeter and 38 km^2 surface area. Port Foster connects to the Bransfield Strait through the

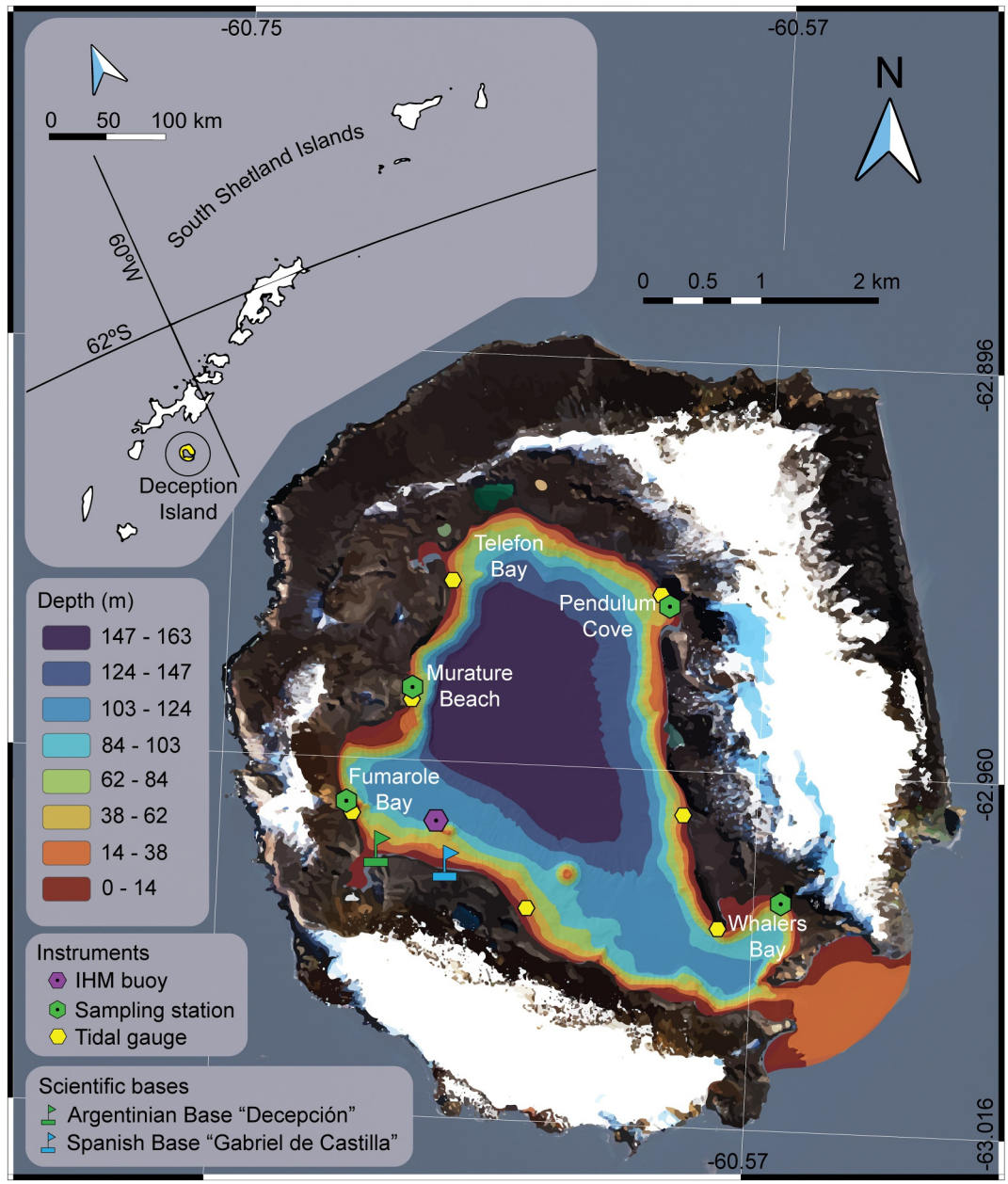


Figure 1. Deception Island close-up view is a Sentinel-2 satellite merged with the colormap bathymetry of Port Foster. Fumarole sites studied in this work (Pendulum Cove, Fumarole Bay, Whalers Bay and Murature Beach) are indicated. Inset in upper left corner: South Shetland Islands archipelago. Coordinates are at UTM Zone 20S. The offshore and Port Foster bathymetric data were supplied by the Instituto Hidrográfico de la Marina (IHM of the Spanish Ministry of Defense).

narrow and shallow Neptune's Bellows, where mean depth is 22 m compared to basin depths exceeding 160 m. The island hosts two Antarctic research stations: the Spanish "Gabriel de Castilla" and the Argentine "Decepción."

Tides in Port Foster are mixed and primarily semidiurnal (Caballero-Martínez, Zarzuelo, et al., 2026). The largest-amplitude component is the principal lunar constituent M_2 (12.42 h). Diurnal amplitudes are lower but comparable in magnitude to those of the semidiurnals. Both the lunar-solar diurnal K_1 (23.93 h) and the principal lunar diurnal O_1 (25.82 h) exceed the second-largest semidiurnal constituent, the principal solar semidiurnal S_2 (12.00 h). The tidal regime is mainly determined by the superposition of these constituents. The superposition of M_2 and S_2 drives a ~14.8 days spring-neap cycle, just as K_1 and O_1 yield a ~13.7 days equatorial-tropic cycle

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Table 1
Sampling Station Locations, Depth (Relative to Mean Sea Level), and Monitoring Intervals

Location	Lat. [S]	Lon. [W]	Depth [m]	Start date	End date
Pendulum Cove	62°56′11″	60°35′33″	3.7	02/21/24	02/29/24
Fumarole Bay	62°58′02″	60°42′34″	2.8	03/01/24	03/06/24
Whalers Bay	62°59′06″	60°34′40″	3.5	03/10/24	03/20/24
Murature Beach	62°56′59″	60°41′24″	∼3.5	01/30/25	02/06/25

Note. The mooring depth at Murature remains approximate, inferred from IHM official nautical charts and water-level time series.

(Valle-Levinson, 2022). When diurnal and semidiurnal constituents are in-phase, larger tidal ranges yield higher high waters (HHW) and lower low waters (LLW). However, when they are out-of-phase, opposing amplitudes reduce the tidal range to higher low waters (HLW) and lower high waters (LHW). This variability is expected to be present in biogeochemical variables in Port Foster.

Port Foster features a deep central basin and a shallow sill at Neptune's Bellows (Hopfenblatt et al., 2021), resembling a fjord. Characterized by weak basin tidal currents and low meltwater inputs (Caballero-Martínez, Zarzuelo, et al., 2026; Torrecillas et al., 2024), it falls within the fjord-type circulation regime of the Geyer and MacCready (2014) estuarine classification framework. According to these authors, estuaries can be classified based on tidal mixing (quantified by a mixing parameter M that depends on characteristic tidal velocity, depth, and buoyancy frequency) and freshwater runoff (parameterized by the freshwater Froude number Fr_f). The $M - Fr_f$ parameter space groups estuaries with similar hydrodynamics. In this parameter space, fjords usually exhibit low M and Fr_f values, as Port Foster does, due to their deep basins, and low characteristic tidal and runoff velocities.

Data from the Spanish Meteorological Agency (AEMET) provided 10-min averaged wind velocities and direction, along with interval peak gusts. The local wind regime is predominantly southwesterly, followed by winds from easterly and northwesterly sectors, with austral summer mean velocities of $\sim$6.2 m s^{-1}. Maximum mean winds occur primarily in the southwest sector, exceeding 25 m s^{-1} with brief gusts up to 40 m s^{-1}.

Several shallow coastal areas around Port Foster, viz. Pendulum Cove, Fumarole Bay, Whalers Bay, and Murature Beach (Figure 1), exhibit intense geothermal and hydrothermal activity. Previous studies (Belyaev et al., 2024; Somoza et al., 2004; Valentin et al., 1989) report interstitial water temperatures near 100 °C alongside magma-driven degassing composed primarily of CO_2 (75–90 %) and methane (CH_4). These emissions create significant biogeochemical anomalies near hydrothermal vents, elevating local concentrations of trace metals (iron, nickel, cobalt, vanadium) and greenhouse gases (CO_2, CH_4). However, despite this intense localized outgassing, Port Foster overall functions as a mild CO_2 sink (Belyaev et al., 2024) during summer conditions. Seasonal variability remains poorly constrained due to a lack of measurements during winter conditions.

2.2. Field Measurements

Data were collected across four deployment periods over a 2-year field campaign. Calibrated sensors were employed to measure different variables: temperature (T), and salinity (S) with a CTD-RBR XR-420, dissolved oxygen (DO) with a MiniDOT logger, and partial pressure of carbon dioxide $\left(p_{CO_2}\right)$ using a SAMI - CO_2. In 2024, these sensors were first installed at Pendulum Cove for about 7 days, then moved to Fumarole Bay for roughly 6.5 days, and finally to Whalers Bay for another 10 days. In 2025, the setup was deployed at Murature Beach for 7 days. Sensor sampling intervals were synchronized to maximize simultaneous measurements. Sensor locations and recording dates are detailed in Figure 1 and Table 1, respectively.

The sampling station incorporated a MiniDOT logger (PME), a CTD-RBR XR-420 (RBR Ltd.), and a SAMI-CO_2 sensor (Sunburst Sensors). The MiniDOT measured dissolved O_2 using an optical fluorescence method, recording DO in mg L^{-1} (later converted to μmol kg^{-1}). It has an accuracy of $\pm$0.3 mg L^{-1}, 0.01 mg L^{-1} resolution, and a 0–150 % saturation range, sampling every 15 min. The MiniDOT optode was factory-calibrated and further verified before the field campaigns through a multi-sensor intercalibration experiment in a seawater tank at

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ICMAN-CSIC (August 2023), including comparisons with independent oxygen sensors and with discrete samples analyzed by Winkler titration. Since a constant salinity of 34 psu and a pressure of 1 atm were set in the instrument configuration before deployment, raw DO values were subsequently corrected using the in situ salinity measured by the co-located CTD and the local atmospheric pressure (mbar), applying the salinity compensation stipulated in the manufacturer's manual (PME), based on the oxygen solubility equations of (García & Gordon, 1992), in line with the processing recommendations of (Bittig et al., 2018). DO was finally converted from $\mathrm{mg\,L^{-1}}$ to $\mu\mathrm{mol\,kg^{-1}}$ using seawater density computed from in situ temperature and salinity.

The CTD-RBR XR-420 measured conductivity, temperature, and pressure at 15 - min intervals. The device has a temperature range from -5℃ to 35℃, with an accuracy of ± 0.002℃ and a resolution below 0.00005℃. Conductivity is measured in the range $0-85\,\mathrm{mS\,cm^{-1}}$, with an accuracy of $\pm 0.003\,\mathrm{mS\,cm^{-1}}$ and a resolution of $1\,\mu\mathrm{S\,cm^{-1}}$. The pressure measurement has a precision of $\pm 0.05\%$ and a resolution below 0.001%. Salinity was derived from the conductivity and temperature measurements. The CTD-RBR XR-420 is factory-calibrated by the manufacturer (RBR Ltd.). Additionally, prior to the field campaigns, its performance was verified in a 4-day intercalibration experiment conducted in a seawater tank at the ICMAN-CSIC facilities (August 2023), where its records were compared with those of an independent reference CTD (SeaBird) and several autonomous temperature-pressure loggers sampling simultaneously. Temperature and pressure differences among sensors were on the order of 10^{-1}℃ and 10^{-1} dbar, respectively, and conductivity records were consistent with the reference CTD, so no additional correction was applied to the CTD-RBR data. Furthermore, upon arrival at Deception Island and before each deployment, all sensors were operated in situ to verify that transport had not affected them and that they functioned correctly.

The SAMI-CO_2 sensor measured p_{CO_2} using a colorimetric reagent method, with an accuracy of $\pm 3\,\mu$atm and $<1\,\mu$atm resolution. The sensor was factory-calibrated by Sunburst Sensors over the range $0-400\,\mu$atm, and its response was further validated up to $\sim 2000\,\mu$atm in laboratory tank experiments prior to the field campaigns, by comparison with p_{CO_2} values calculated from discrete seawater samples (pH and total alkalinity). Beyond $\sim 2000\,\mu$atm, reduced indicator sensitivity increases measurement uncertainty due to calibration extrapolation (DeGrandpre et al., 1995). Consequently, values exceeding this threshold are considered semi-quantitative, order-of-magnitude estimates of hydrothermal CO_2 enrichment. The SAMI-CO_2 determines p_{CO_2} from the absorbance of a bromothymol blue indicator solution enclosed in a gas-permeable membrane (DeGrandpre et al., 1995). The sensor and its reagent were factory-calibrated by Sunburst Sensors for low-temperature operation (seawater temperatures below 5℃), representative of the environmental conditions of the deployment site, where seawater temperature never dropped below 0℃ during the field campaigns. Sampling was programmed to collect p_{CO_2} data every 30 min.

At each location, SAMI, MiniDOT and CTD-RBR recorded measurements at ~ 0.5 m below the surface. The locations of the sampling stations were moored as close to the shore as possible, within the zone where hydrothermal vents are present, and consistently kept below the lowest tidal level.

Water level variations were recorded with pressure sensors (Aqualogger PT520 model from AQUATEC) moored around the perimeter of Port Foster at depths between 2.8 m and 4 m (Table 1). Data consisted of 3 - min averages from bursts of six measurements every 10 s. Pressure was converted to surface elevation following Nielsen (1989) assuming a constant density of $1027.3\,\mathrm{kg\,m^{-3}}$, as thermohaline-induced density variations were negligible. Depths were initially measured using a handheld depth finder, yielding values of 3.7 m, 2.8 m, and 3.5 m for Pendulum Cove, Fumarole Bay, and Whalers Bay, respectively. However, the depth finder stopped working during the 2025 campaign, so the mooring depth at Murature was estimated at approximately 3.5 m based on measured water level variations.

Wave parameters (mean, significant, and maximum heights and periods) were recorded every 30 min from 01/20/24 to 03/20/24 by an oceanographic buoy from the Spanish Navy Hydrographic Institute (IHM), located off the Spanish base "Gabriel de Castilla."

Wind velocity and direction were obtained from the WMO-89070 meteorological station at "Gabriel de Castilla" (62 °58′38″ S, 60 °40′33″ W), part of the Antarctic meteorological network installed and managed by the Spanish Meteorological Agency (AEMET; Agencia Estatal de Meteorología and Buisan (2025)), providing

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10 - min measurements at 10 m height, including temperature, humidity, net incident solar radiation, precipitation and pressure.

Discrete surface seawater samples for biogeochemical analysis were hand-collected at the mooring sites. Total alkalinity (A_T, μmol kg^{-1}), pH in Total Scale (at 25°C, pH_{T25}) and chlorophyll-a (Chl-*a*) were measured. Samples of A_T were poisoned with 100 μL of $HgCl_2$ and determined (twice per sample) using a Metrohm Titroprocessor 888, following Mintrop et al. (2000), with precision ($\pm$ 2, 163 μmol kg^{-1}) and accuracy ($\pm$ 3.25 μmol kg^{-1}) being verified by measurements of CRM batch #207 provided by the Scripps Institution of Oceanography. pH was measured in optical glass cells stabilized at 25°C using a Shimadzu UV-2401PC spectrophotometer, following Clayton and Byrne (1993). Precision ($\pm$ 0.0026) and accuracy ($\pm$, 0.0071) were similarly verified using CRM batch #207. For biogeochemical interpretation of the high frequency data, p_{CO_2} and dissolved inorganic carbon (DIC) were calculated using the pair carbonate system parameters (A_T-pH) with the CO2SYS software for MATLABTM (Sharp et al., 2020; Van Heuven et al., 2011), incorporating nutrients (not shown) and dissociation constants from Waters et al. (2014), developed for a temperature range of 0−50°C and salinity of 1−50 for real seawater. The borate-salinity ratio from Lee et al. (2010), the sulfate constant defined by Dickson et al. (2007) and the fluoride constant from Perez and Fraga (1987) were also used. The uncertainty error propagation was calculated, resulting in 10.3 ppm for p_{CO_2} (Blasco & Tovar-Sánchez, 2023). Chl-*a* concentration was obtained by filtering known volumes of seawater (500 mL) through a Whatman 0.7 μm GF/F glass fiber filter that was maintained frozen in the dark until analysis. After defrosting, 90 % acetone was added as an extractant and measured by fluorometry following Amaya-Vías et al. (2025). The estimated error in the measurements was $\pm$0.5 μg L^{-1}.

Drone flights acquired high-resolution thermal imagery to support the characterization of the study sites. Because hydrothermal vents are visually identifiable via fumarole emissions (Valentin et al., 1989), both moorings and flight paths were planned prior to the field campaign. One overflight per site was conducted on 29 January 2025 (Whalers Bay, Pendulum Cove) and 30 January 2025 (Fumarole Bay), complying with Spanish Agency for Aviation Safety (AESA) regulations and SCAR environmental guidelines (SCAR, 2017). Two platforms were used: the DJI Mavic 2 Enterprise Advanced (M2EA), equipped with a 640 × 512 px Uncooled VOx Microbolometer, and the DJI Matrice 300 RTK (M300) with the DJI Zenmuse H20T thermal sensor (13.5 mm focal length). Flights were planned with the DJI Pilot App to fully cover each area, using a 50 m altitude and 80−70% frontal and side overlap for photogrammetric processing. RAW images were first converted to radiometric TIFF format using ATygeo (accounting for humidity, reflection angle, and flight altitude), and then processed in Pix4D Mapper (v.4.8.3). The processing chain included image alignment, generation of a 3D dense cloud, and the creation of a Digital Surface Model and a thermal orthomosaic representing relative temperature (°C).

Finally, atmospheric CO_2 levels during the field campaign were obtained (in ppm) from the National Oceanic and Atmospheric Administration global monitoring network (NOAA) located in Palmer Station (Lan et al., 2025). All dates follow the mm/dd/yy format.

2.3. Air–Sea CO_2 Fluxes Calculation

In order to calculate the CO_2 exchange between sea and atmosphere, several environmental factors were considered: p_{CO_2} in water, wet p_{CO_2} in air, wind velocity at 10 m reference height, salinity and temperature. The air–sea CO_2 flux $F(CO_2)$ (in mmol m^{-2} d^{-1}, i.e., mass per unit area and unit time) was calculated according to Wanninkhof (2014) as

$$F(CO_2) = k \cdot k_0 \cdot \left(p_{CO_{2\,water}} - p_{CO_{2\,air}}\right), \quad (1)$$

where k (typically in cm h^{-1}) represents the gas transfer velocity and k_0 (in mol m^{-3} atm^{-1}) is the aqueous phase solubility of CO_2 at in situ temperature and salinity, as described by R. Weiss (1974). Furthermore, atmospheric p_{CO_2} values were adjusted to wet conditions following the methodology of R. F. Weiss and Price (1980), using locally measured atmospheric pressure data and considering only the first virial coefficient (R. Weiss, 1974).

To estimate k, the specific characteristics (temperature and salinity) of the study area were considered. The value of k is defined by:

$$k = k_{660} \cdot (S_{cw}/660)^{-1/2}. \quad (2)$$

In Equation 2, the Schmidt number for seawater (S_{cw}) was obtained using the coefficients from Wanninkhof (2014), which were derived from the kinematic viscosity of freshwater and seawater by Sharqawy et al. (2010) and CO_2 diffusion coefficients measured by Jähne et al. (1987). The $-1/2$ exponent in Equation 2 assumes the presence of wind-waves on the air–sea interface (Coantic, 1986; Ledwell, 1984; Wanninkhof, 1992).

Although k is mainly defined using 10 m wind velocity in open ocean waters, where wind stress dominates air–sea turbulence (Wanninkhof, 2014), standard global flux formulations may be inadequate for local environments. In estuarine environments such as Port Foster, sheltered areas shielded from strong winds, fetch-limited wave spectrum, and tidal currents can significantly enhance gas transfer (Jiang et al., 2008; Raymond & Cole, 2001). The approach of Borges et al. (2004) for the gas transfer velocity observed in fjords with unbinned data was adopted (Equation 2). This relationship reads

$$k_{600} = (1.2 + 2.30\, U_{10}), \quad (3)$$

where U_{10} represents the horizontal wind velocity at 10 m height above the sea level. Wind data from "Gabriel de Castilla" Meteorological Station in Deception Island were considered to calculate the fluxes. The corresponding correction of k_{600} to k_{660}, as described by Ho and Wanninkhof (2016), was applied. The application of Equation 3 and Equation 2 together with Equation 1 is particularly appropriate for Port Foster, as it is classified as a fjord-type according to the estuarine parameter space proposed by Geyer and MacCready (2014) (see Section 2.1). The only location where its applicability might be questionable is Neptune's Bellows (Figure 1), where tidal currents are stronger. However, all analyzed stations are located far from the inlet. Figure S1 in Supporting Information S1 compared the Borges et al. (2004) formulation, the Wanninkhof (2014) equation, and other open-sea parameterizations (Ho et al., 2006; Jiang et al., 2008; Nightingale, Liss, & Schlosser, 2000; Nightingale, Malin, et al., 2000). This shows that open-sea formulations tend to overestimate computed air–sea fluxes during strong wind conditions and underestimate them during weak wind events in estuarine and semi-enclosed environments.

2.4. Data Analysis Methods

Joint variability between seawater p_{CO_2}, water temperature (T), salinity (S), sea surface elevation (η), wind velocity (W), wind direction (Dir_W), significant wave height (H_s), and dissolved oxygen (DO) is evaluated. Wavelet analysis was used to investigate time–frequency relationships not readily apparent in the raw records. Wavelet coherence quantified shared variability and phase relationships, while cross-wavelet power identified coincident high-energy fluctuations. Both methods followed Grinsted et al. (2004). A Morlet-type wavelet function, with a dimensionless frequency of 6 and a resolution of 12 scales per octave (1/12), was selected for its suitability for biogeochemical time series (Grinsted et al., 2004). The analyses were restricted to periods with simultaneous data at each location (Table 1). Minimum levels in the time series were detected via changes in the mean.

All data sets and code used to perform the different analyses presented in this study, including data processing, wavelet analyses, and model implementations, can be found in the data repository described in Caballero-Martínez, Amaya-Vías et al. (2026).

3. Results

3.1. Spatio-Temporal Variability of Seawater p_{CO_2}

3.1.1. Time Series of p_{CO_2}

Figure 2 shows the p_{CO_2} time series at Pendulum Cove, Fumarole Bay, Whalers Bay and Murature Beach. All sites exhibit positive exceedances over a background value of p_{CO_2}. Background level refers to the mean seawater p_{CO_2} over which exceedances occur at a given coastal site due to hydrothermal activity. The background level is a site-specific feature: each hydrothermal vent has a different but remarkably close background level. Pendulum Cove displays the smallest peaks above the background level, reaching approximately $1,500\,\mu$atm (panel a). Fumarole Bay and Whalers Bay exhibit more frequent fluctuations than Pendulum Cove, recording intermediate values of $\sim 7,000\,\mu$atm (panel b) and $\sim 2,100\,\mu$atm (panel c), respectively. Murature Beach shows the highest p_{CO_2} pulses,

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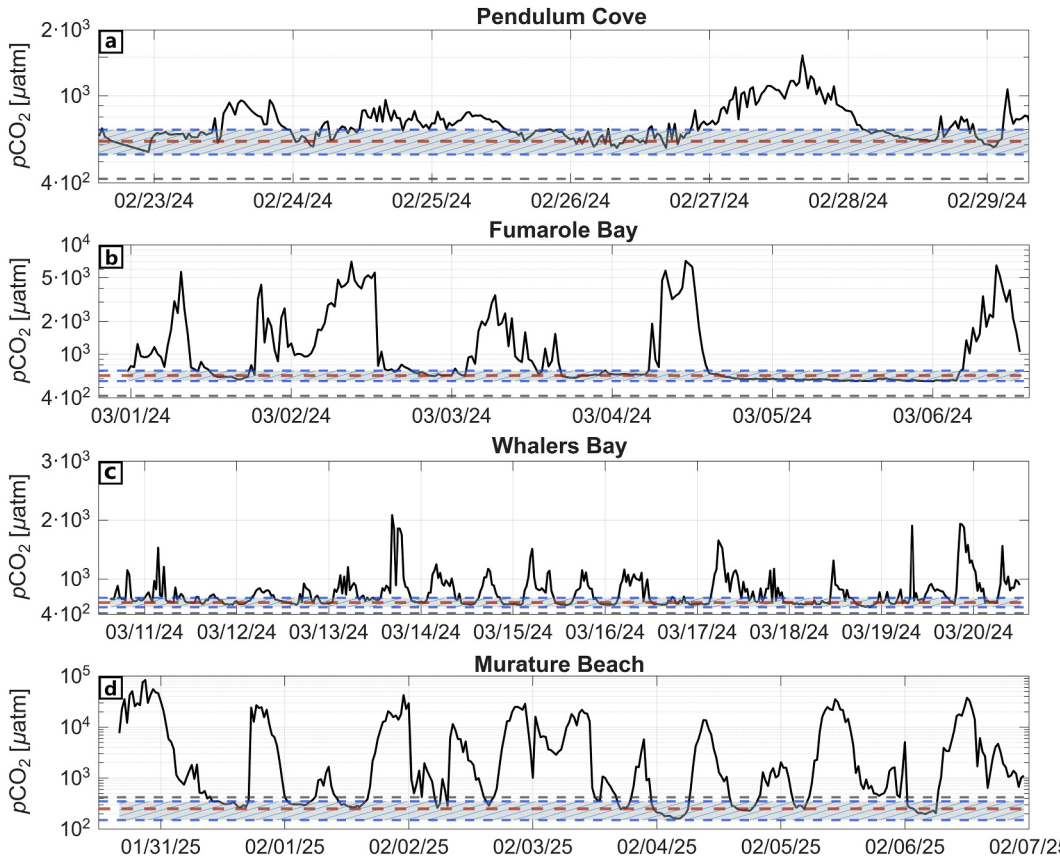


Figure 2. Time series of seawater p_{CO_2} recorded at (a) Pendulum Cove, (b) Fumarole Bay, (c) Whalers Bay, and (d) Murature Beach. Red dashed lines and shaded blue band represent the mean background level and its uncertainty, respectively, for each site. Gray dashed lines indicate for reference monthly averaged air CO_2 concentrations from Palmer Station, in Anver Island. Notice the different horizontal axis for each panel (measurements were not simultaneous). The *y*-axes are presented on a logarithmic scale to make visible background levels.

exceeding 80,000 μatm (panel d), an order of magnitude above other sites. Peaks exceeding the SAMI-CO_2 validated range ($\sim$2,000 μatm; Section 2.2) should be interpreted as semi-quantitative, order-of-magnitude estimates.

Each site exhibits a distinct mean background level (red dashed line, Figure 2) and an associated range of variability (blue shaded band). This range is determined from the time-series minimum to the 50th percentile, except at Murature Beach, where the 30th percentile was used due to its higher p_{CO_2} variability. These levels at Pendulum Cove, Fumarole Bay, and Whalers Bay are strikingly similar between each other, with approximate values of $p_{CO_2b} = (620 \pm 80)\,\mu$atm, $p_{CO_2b} = (640 \pm 70)\,\mu$atm, and $p_{CO_2b} = (600 \pm 80)\,\mu$atm, respectively. Seawater p_{CO_2} consistently exceeds atmospheric CO_2 concentration across all three sites (derived from $\sim$418 ppm dry air following R. F. Weiss and Price (1980), corresponding to $401 - 410\,\mu$atm at Pendulum Cove, $406 - 417\,\mu$atm at Fumarole Bay, and $397 - 408\,\mu$atm at Whalers Bay). In contrast, despite exhibiting the strongest p_{CO_2} pulses, Murature Beach features a lower background level $\left(p_{CO_2b} = 300 \pm 100\,\mu\text{atm}\right)$, remaining below atmospheric concentration ($\sim$421 ppm; $404 - 409\,\mu$atm).

3.1.2. Variability at Tidal Scales

Figure 3 shows wavelet coherence and cross-correlation spectra between time series of temperature, salinity, p_{CO_2}, DO concentration and η at Whalers Bay. Pendulum Cove, Fumarole Bay, and Murature Beach wavelet coherence and cross-correlation results are shown in Supporting Information S1 (Figure S2–S4 in Supporting Information S1, respectively). Paired time series for the analyzed variables (panels a1–e1) display their temporal

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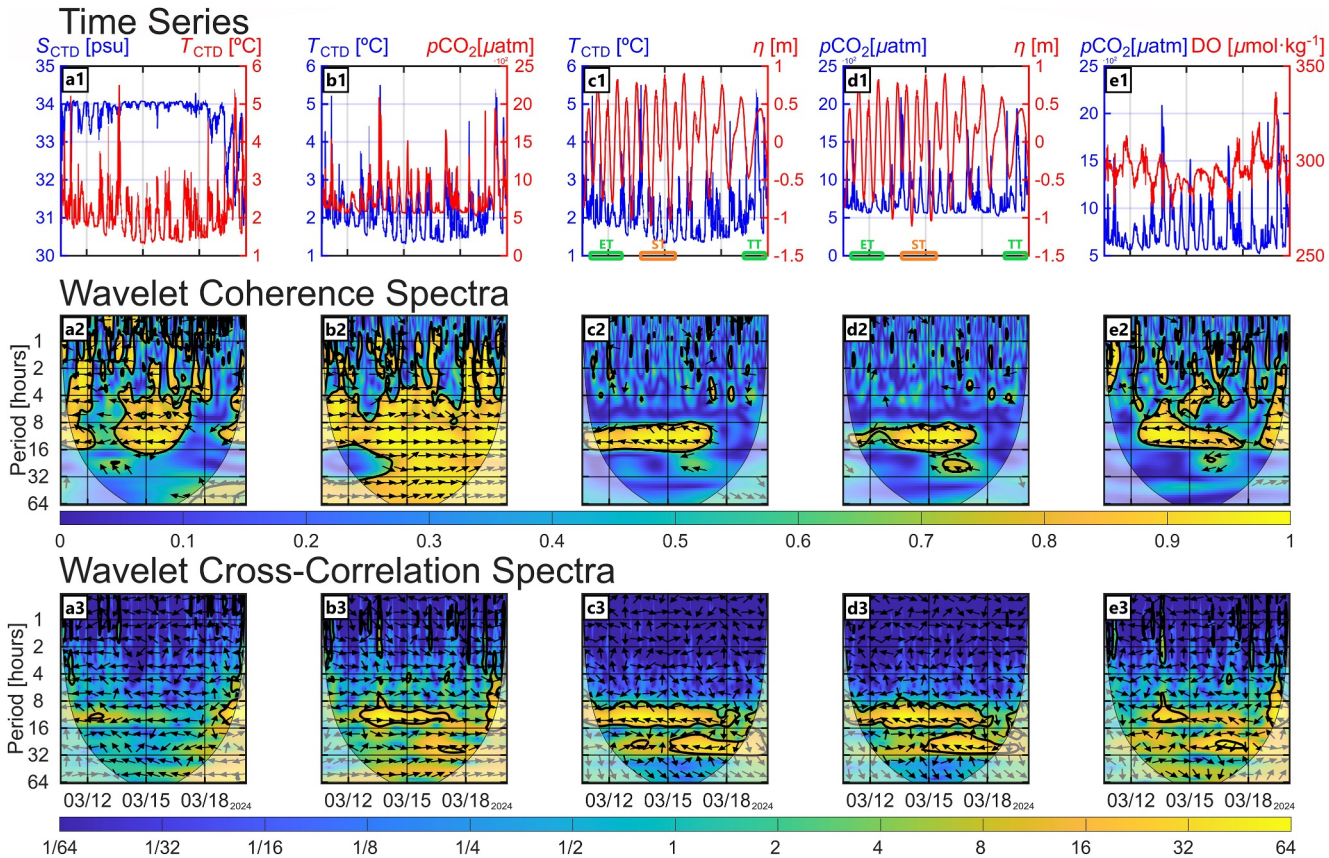


Figure 3. Panels (a1, a2, and a3) show: time series of near-surface sea water salinity (blue curve) and temperature (red curve) near-surface at Whalers Bay, a color plot of their wavelet coherence spectrum and a color plot of their wavelet cross-correlation spectrum, respectively. Panels (b1, b2, and b3): As in panels (a1, a2, and a3), except for temperature (blue curve) and p_{CO_2} (blue curve). Panels (c1, c2, and c3): As in panels (a1, a2, and a3), except for temperature (blue curve) and free-surface elevation (red curve). Panels (d1, d2, and d3): As in panels (a1, a2, and a3), except for p_{CO_2} (blue curve) and free-surface elevation (red curve). Panels (e1, e2, and e3): As in panels (a1, a2, and a3), except for p_{CO_2} (blue curve) and DO (red curve). Warm colors indicate high coherence and high common power in the wavelet coherence and cross-correlation spectra, respectively. Thick black lines encircle 95% statistically significant regions of the spectra. Arrows indicate phase lags between the time series, viz. in-phase if pointing right, anti-phase if pointing left, and +90°-phase lag leading the first variable if pointing downward. Shaded areas indicate where edge effects might distort the spectra. Green and orange rectangles along the *x*-axis of panels (c1 and d1) highlight tidal phases in η time series: ST, spring tides; ET, equatorial tides; and TT, tropic tides.

evolution. Wavelet coherence spectrum (panels a2–e2) quantifies time-frequency phase synchronization between the two time series, separating uncoupled periods (values near zero) from strong linear coupling (values close to 1). Cross-wavelet power spectra (panels a3–e3) isolate coincident high-energy periods, where high values define simultaneous high-power fluctuations in both time series and values near 0 represent uncoupled variability.

The first column of Figure 3 illustrates the temperature-salinity relationship. Panel a1 displays the temporal variability of both parameters across the observation period. At Whalers Bay, the wavelet coherence spectrum (panel a2) shows strong co-variability, with coherence higher than 0.8, spanning from the shortest resolved timescales of a few hours up to the tidal frequencies (semidiurnal and diurnal periods). Similar high-coherence regions appear at Fumarole Bay and Murature at tidal and longer periods. Pendulum Cove (Figure S2 in Supporting Information S1) exhibits even stronger coherence (~1) across almost all periods exceeding 3 h, suggesting that temperature-salinity variability is driven by both tides and additional processes like hydrothermal inputs (detailed in Sections 3.1.3 and discussed in Section 4.1.3). At all four sites, the phase arrows point left, indicating an anti-phase relationship: increases in temperature are coherent with decreases in salinity.

Power cross-correlation between temperature and salinity time series is weak at all four sites (as shown at Figures S2–S4 in Supporting Information S1), except at semidiurnal periods on 03/12/24, as well as on 03/19/24 at Whalers Bay. Individual wavelet transforms (not shown) also display low energy outside these specific intervals,

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suggesting that variability during these specific periods is driven by tides. As for the wavelet coherence spectrum (panel a2), phase lags in significant parts of the cross-correlation spectrum (panel a3) are consistently in anti-phase. This indicates again that temperature and salinity are anti-correlated at semidiurnal tidal periods.

Wavelet coherence between the water temperature and p_{CO_2} time series (panel b1) is overall high in Whalers Bay (panel b2) and Fumarole Bay (Figure S3 in Supporting Information S1). In other sites the coherence is generally weaker. At Pendulum Cove, wavelet coherence values are below 0.5 for most of the monitoring interval, suggesting that the two series vary mostly independently. In contrast, Murature Beach shows certain periods of strong coherence (above 0.8), with both time series varying synchronously at tidal frequencies (e.g., during 02/02/25 and 02/03/25). However, other time intervals show weak coupling (e.g., from 02/04/25 to 02/06/25). Phase lags in all statistically significant regions indicate that temperature and p_{CO_2} are approximately in-phase (arrows pointing right). This water temperature$-p_{CO_2}$ relationship may reflect hydrothermal degassing and associated thermal anomalies (Kelley et al., 2001). Panel b3 reveals significant temperature$-p_{CO_2}$ power cross-correlation only during brief intervals at tidal periods (~12 h) at Whalers Bay, Fumarole Bay, and Murature Beach. No significant cross-correlations are obtained at Pendulum Cove. No common power is found at other frequencies.

The influence of equatorial tides is observed in the temperature$-\eta$ coherence spectrum, where correlation values approach 1 at semidiurnal frequencies during equatorial tides (panel c2), when diurnal inequality is at its lowest (panel c1). High values of coherence and cross-wavelet power also emerge in the semidiurnal band during equatorial tides at Pendulum Cove and Murature Beach (Figures S2 and S4 in Supporting Information S1). Conversely, peak diurnal inequality during tropic tides shifts coherence and cross-wavelet power to diurnal frequencies. This is particularly evident at Fumarole Bay (see Figure S3 in Supporting Information S1). Overall, wavelet analysis shows a strong influence of tides on temperature variability, which was not evident from direct observation of temperature time series.

Phase lags in the temperature$-\eta$ spectra are close to 180° (panels c1–c3), indicating an anti-phase relationship at semidiurnal and diurnal periods (~12 to ~24 h). When temperature increases, η decreases, and vice versa. At tidal scales, at HHW or LHW, temperature attains relative lower values, whereas at LLW or HLW, temperature attains the highest values. Figure 3 excludes salinity$-\eta$ time series and wavelet analyses for readability. These results are provided in Supporting Information S1 for all sites (Figures S2–S5 in Supporting Information S1, panels f1–f3 for Pendulum Cove, Fumarole Bay, Murature Beach, and Whalers Bay, respectively). Unlike temperature$-\eta$, the salinity$-\eta$ time series (panels f1–f3 in Figure S5 in Supporting Information S1) are near in-phase, reflecting the negative temperature-salinity correlation (panels a1–a3 from Figure 3). During high water levels (HHW and LHW) the saltier water mass in Port Foster mixes with that of the coastal site due to the presence of freshwater outflow from glacier melting. However, for the same reason, at low water levels (LLW and HLW) salinity decreases and higher temperatures are observed. Salinity$-\eta$ wavelet analysis shows high coherence (above 0.8) at semidiurnal and diurnal tidal bands, although the co-variability is much weaker at Murature Beach (panel f2 in Figure S4 in Supporting Information S1).

Wavelet coherence between η and p_{CO_2} (panel d1) is close to 1 at semidiurnal tidal band (panel d2) with a noticeable 180° phase lag, that is, anti-phase. Fumarole Bay exhibits diurnal coherence during tropical tides, while Murature Beach shows in-phase co-variability across both diurnal and semidiurnal bands (Figures S3 and S4 in Supporting Information S1, respectively). However, no significant coherence is obtained at Pendulum Cove (Figure S2 in Supporting Information S1). Similar to temperature and salinity, persistent statistically significant cross-correlation (panel d3) between η and p_{CO_2} emerged at semidiurnal frequencies, with 180° phase lag between them.

At Whalers Bay, p_{CO_2} and DO exhibit an inverse relationship (panel e1) featuring a 180° phase lag in both coherence and cross-correlation spectra (panels e2 and e3). Fumarole Bay and Murature Beach (Figures S3 and S4 in Supporting Information S1) display similar tidally-driven variability. No clear p_{CO_2}−DO relationship is observed at Pendulum Cove, suggesting additional drivers of variability.

In summary, although there are site-specific differences, both tidal semidiurnal and diurnal control prevails at all shallow-water hydrothermal vents in Port Foster. Overall, low tidal water levels correspond with warmer, less saline and p_{CO_2} saturated waters, whereas high tidal water levels correspond to colder, saltier, and less p_{CO_2} saturated waters.

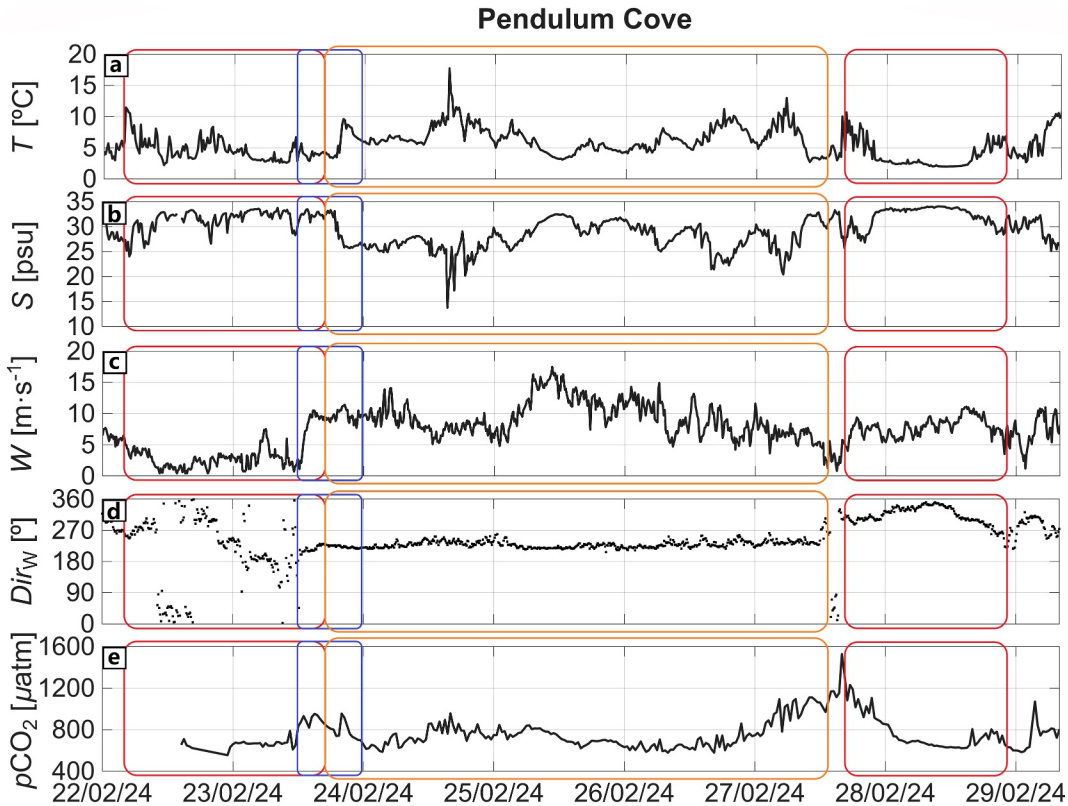


Figure 4. Time series of: seawater temperature (panel a), seawater salinity (panel b), wind velocity (panel c), wind direction (0° and 90° are winds blowing from North and East, respectively; panel d) and p_{CO_2} (panel e) at Pendulum Cove. Colored rectangles highlight specific event windows: red indicate period in which most of the temporal variability of salinity, temperature and p_{CO_2} is suppressed under varying wind directions, blue denotes an increase in water temperature paired with a drop in salinity on 02/23/24, and orange show a period of persistent WSW wind conditions.

3.1.3. Variability at Event Scales

Event-scale processes (i.e., short-lived changes occurring over hours to a few days, typically associated with rapid meteorological shifts) also drive variability in water biogeochemistry. For example, at Pendulum Cove, on 02/23/24, temperature increased by ~6°C and salinity decreased ~7.2 psu within 1 hr 45 min (identified by a blue rectangle in Figure 4), coincident with intensified winds veering from ESE to WSW. Winds from the WNW direction lead to a decrease in water temperature and an increase in salinity. This pattern is observed during both 02/22/24–02/23/24 and 02/28/24 (red rectangles in Figure 4). From 02/24/24 to 02/27/24, when a period of sustained WSW winds occurred following a wind shift from the ESE (highlighted by the orange rectangle), seawater temperature decreased as wind velocity increased. The observations suggest that the shifts in temperature and salinity are produced by advection of the temperature and salinity gradients, respectively. North and west of the mooring site, salinity and temperature are lower due to the presence of freshwater outflow from glacier melting. The thermal imagery acquired in 2025 at Pendulum Cove (Figure 5a) provides independent spatial evidence of the location of this freshwater input, although it was not contemporaneous with the 2024 event. Panel e further illustrates p_{CO_2} variations driven by other forcings, such as WNW wind events inducing a decrease of seawater p_{CO_2} (red rectangles).

Figure 6 illustrates the influence of wind and wind-waves on temperature, salinity, and p_{CO_2} at Fumarole Bay, the site closest to the IHM wave buoy. On 03/05/24, storm-driven onshore winds and high waves (purple rectangles, panels a–f) intensified the mixing of the water column and wave dissipation, driving CO_2 dispersion and exchange with Port Foster. The effects of mixing can be observed in temperature (panel a), salinity (panel b) and p_{CO_2} (panel f), whose values remain almost constant during that day. Fumarole Bay and Whalers Bay might not be affected by changes in water temperature due to the absence of significant discharge of freshwater from glacier

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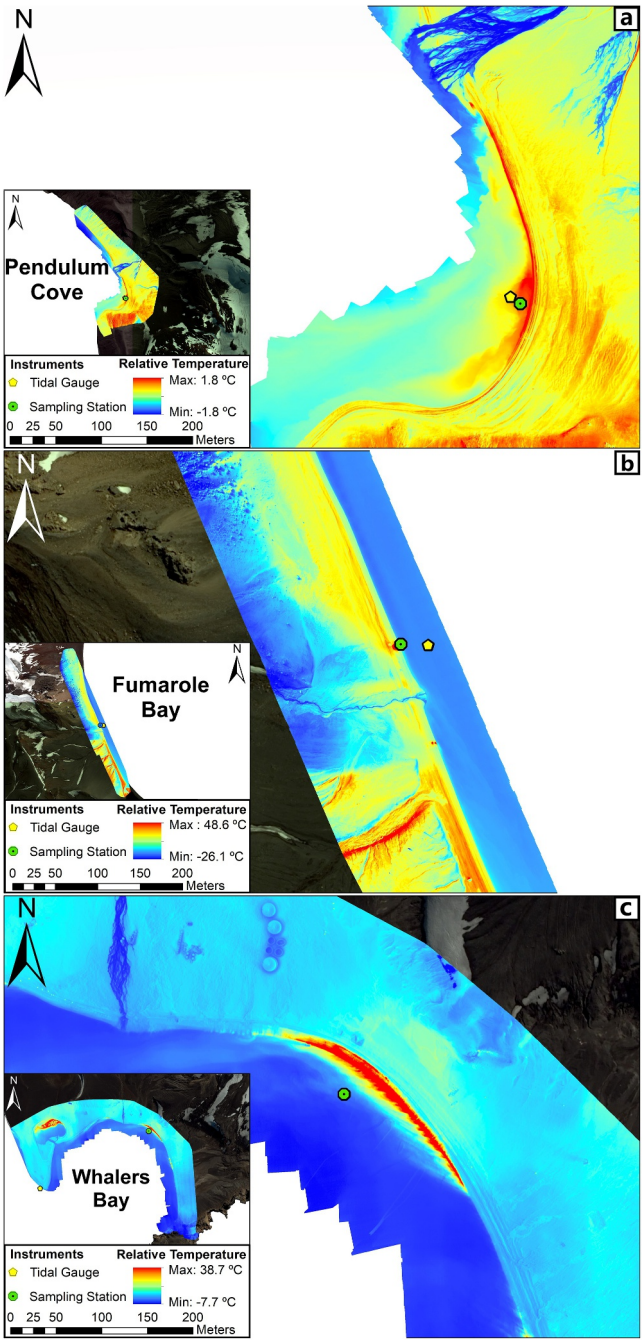


Figure 5. Thermal footprints acquired via drone flights using a DJI Matrice 300 RTK equipped with a Zenmuse H20T thermal sensor over three different sites in Port Foster, Deception Island, on 29 January 2025 at Whalers Bay and Pendulum Cove, and on 30 January 2025 at Fumarole Bay, during the field campaign of the DICHOSO project. Panel (a) corresponds with Pendulum Cove, panel (b) with Fumarole Bay and panel (c) with Whalers Bay. Color scales and moorings are in the respective legends.

melting (Figures 5b and 5c). Unlike the pre-storm conditions, p_{CO_2} showed no significant variations during the storm, remaining close to background levels (panel f). Moreover, temperature and salinity exhibit local maxima and minima during transitions between high and low water levels.

3.2. Local Air–Sea CO_2 Fluxes

The minimum and maximum of the observed fluxes ($F(CO_2)$; red and blue solid curves in Figure 7) are: $4-134\,\mathrm{mmol\,m^{-2}\,d^{-1}}$ at Pendulum Cove (panel a), $3-529\,\mathrm{mmol\,m^{-2}\,d^{-1}}$ at Fumarole Bay (panel b), and $2-228\,\mathrm{mmol\,m^{-2}\,d^{-1}}$ at Whalers Bay (panel c). At Murature Beach, $F(CO_2)$ time series ranges from -13 to $11,260\,\mathrm{mmol\,m^{-2}\,d^{-1}}$ (panel d). Murature Beach exhibits both the only negative fluxes (temporary CO_2 uptake) observed in Port Foster and the highest emissions, reaching values approximately one to two orders of magnitude higher than at the other sites, depending on the site considered. Negative fluxes occurred when seawater p_{CO_2} fell below atmospheric levels, likely reflecting net autotrophic conditions where photosynthetic CO_2 uptake exceeds respiration and remineralization (blue curve in panel d). This interpretation is supported by elevated DO levels, the inverse $p_{CO_2}-$DO relationship, and high Chl-*a* concentrations (27–60 $\mu g\,L^{-1}$) measured at Murature Beach in February 2025. This yielded an intense phytoplankton bloom occurring at Murature Beach during this period.

The temporal variability of air–sea fluxes is similar to that of the seawater p_{CO_2}. Air–sea fluxes have a stronger dependence on p_{CO_2} than on temperature, salinity, or wind. A sensitivity analysis was performed by comparing observed fluxes with scenarios where each parameter was held constant at its minimum and maximum observed values. The results of the sensitivity analysis are shown in Figure S6 in Supporting Information S1. The results indicate that the wind velocity markedly influences flux magnitude. However, the overall co-variability and temporal pattern remain consistent across all cases. Observed fluxes show minimal sensitivity to variations in temperature and salinity within the measured range. This suggests that the air–sea p_{CO_2} gradient acts as the main factor that governs the temporal co-variability of CO_2 fluxes, whereas wind velocity regulates the gas transfer velocity (k), thereby controlling their overall magnitude.

4. Discussion

4.1. Spatio-Temporal Joint Variability

4.1.1. p_{CO_2} Temporal Variability Due To Biogeochemical Processes

A key feature of p_{CO_2} temporal variability at fumarolic sites at coastal waters of Deception Island is the presence of regular fluctuations above site-specific background levels. Pendulum Cove, Fumarole Bay, and Whalers Bay show strikingly similar background levels ($600-640\,\mu$atm), suggesting shared ambient conditions. These values remain above the atmospheric CO_2 levels ($397-410\,\mu$atm). Chl-*a* concentrations measured at these sites during the surveys in 2024 were consistently low, with average values of $0.52\pm0.22\,\mu g\,L^{-1}$ ($n=26$) at Pendulum Cove, $0.66\pm0.15\,\mu g\,L^{-1}$ ($n=10$) at Fumarole Bay and $0.732\pm0.25\,\mu g\,L^{-1}$ ($n=31$) at Whalers Bay. These concentrations indicate low phytoplankton biomass and suggest that primary production had limited influence on background p_{CO_2} levels in relation to hydrothermal inputs and physical mixing. In contrast, the $300\,\mu$atm background level at Murature Beach in February 2025 was roughly half that of the other sites, showing values generally below the atmospheric

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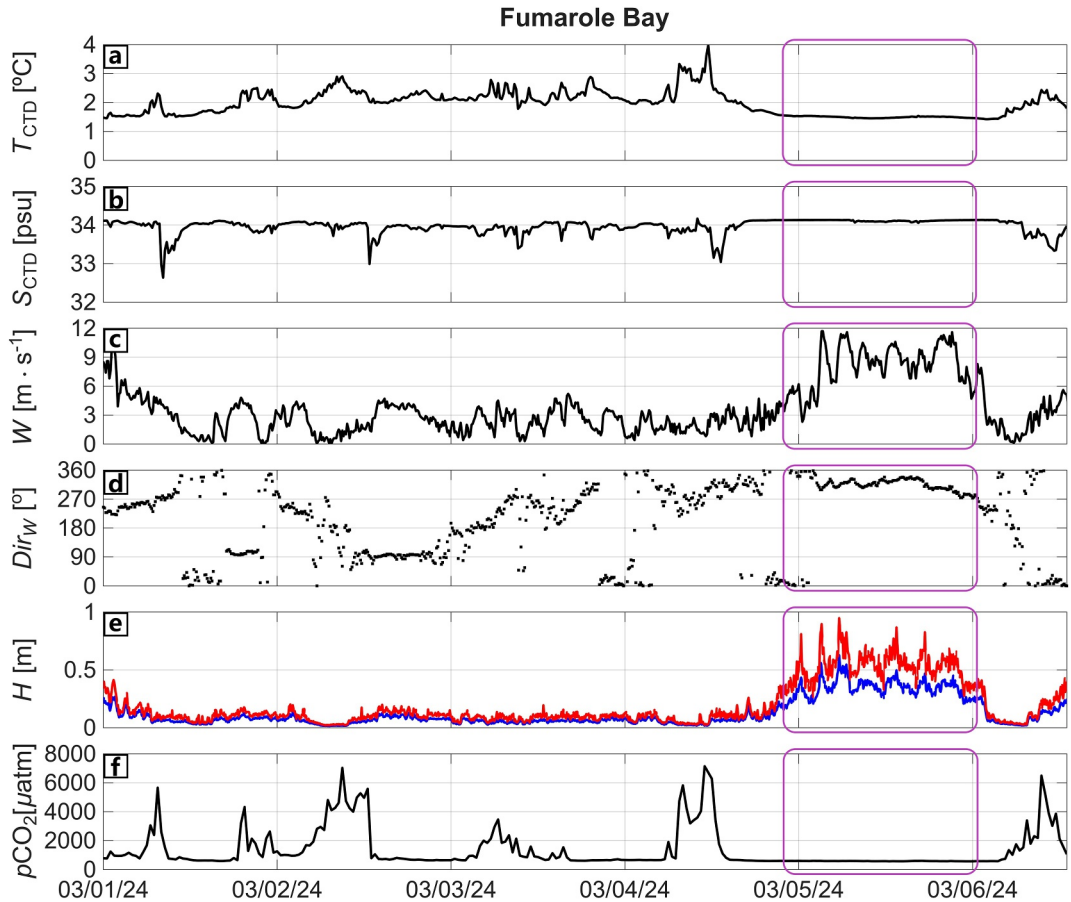


Figure 6. Time series at Fumarole Bay showing seawater temperature in panel a, seawater salinity in panel (b), wind velocity in panel (c), wind direction in panel (d), maximum (red line) and significant (blue line) wave height in panel (e), and p_{CO_2} in panel (f). Purple rectangle (panels a–f) highlights a storm event on 03/05/24.

levels (404−409 μatm) This difference may be partly explained by enhanced biological CO_2 uptake during an intense phytoplankton bloom, evidenced by high average Chl-*a* concentrations of 44 ± 19 μg L^{-1} (n = 15). The associated photosynthetic consumption of CO_2 and production of oxygen align with elevated DO levels (panel e1 in Figure S4 in Supporting Information S1) and the strong inverse p_{CO2}−DO relationship observed (panel e2 of Figure S4 in Supporting Information S1). As a first-order feasibility assessment, carbonate-system calculations based on the thermohaline and carbonate-system variables measured at Murature Beach ($A_T = 2,485$ μmol kg^{-1} and $pH_{T25} = 7,5853$) indicate that a decrease in p_{CO_2} from approximately 640 μatm to 300 μatm would correspond to a DIC reduction of approximately 77 μmol kg^{-1}. Such a drawdown appears plausible under the high-phytoplankton-biomass conditions revealed by the high Chl-*a* concentration measured in this coastal site.

Hence, background p_{CO_2} values primarily reflect the properties of the Port Foster main water mass, modified locally by hydrothermal inputs, physical mixing, and, at Murature Beach, biological CO_2 uptake. The highest background level occurs at Fumarole Bay $\left(p_{CO_2b} = 640 \pm 70\,\mu\text{atm}\right)$. This localized geothermal outgassing establishes a pronounced spatial p_{CO_2} gradient between the shallow hydrothermal vents and the deeper main basin of Port Foster. Driven by intense physical mixing from wind and tidal action, horizontal dispersive transport of CO_2 dominates over coastal advection. At Pendulum Cove, Fumarole Bay, and Whalers Bay, minimum p_{CO_2} nearshore concentrations exceed those in the central basin of Port Foster (∼500−650 μatm) according to Amaya-Vías et al. (2026). SOCAT database (version 2025; Bakker et al. (2016, 2025)) also presents p_{CO_2} data from different cruises inside Port Foster on single days from 2000 to 2023, with values from 223 μatm to 583 μatm.

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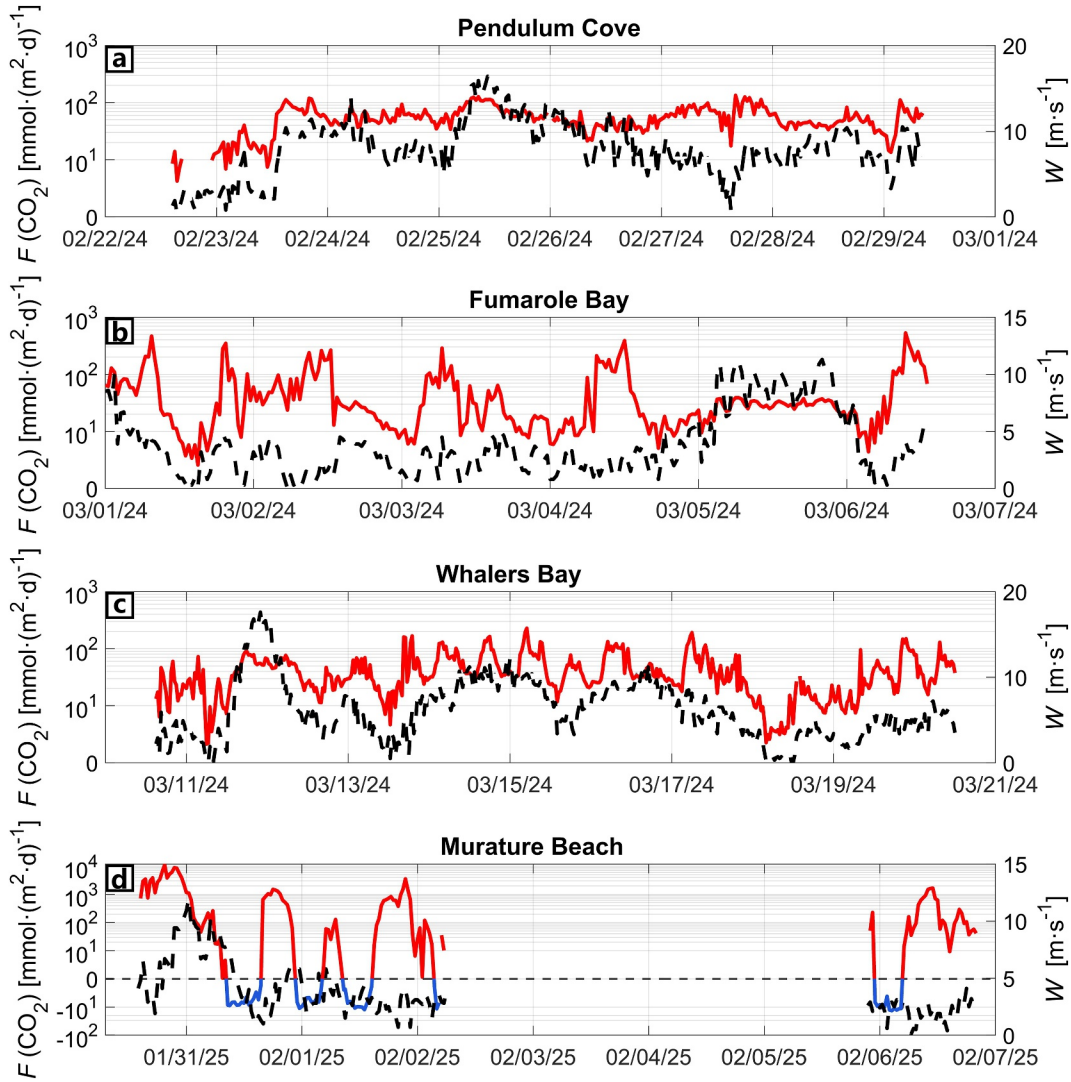


Figure 7. Time series of observed air–sea flux (solid curve) and wind measurements (black dashed curve) at Pendulum Cove (panel a), Fumarole Bay (panel b), Whalers Bay (panel c), and Murature Beach (panel d). Red (Blue) solid curve indicates positive (negative) air–sea flux values. Left *y*-axis is plotted on a logarithmic scale for air–sea curve, while, while right *y*-axis is plotted on a linear scale for wind data.

Background p_{CO_2} levels remain far below the pulse peaks measured at each site. During these extreme events, the massive and rapid hydrothermal CO_2 supply overwhelms phytoplanktonic uptake capacity to substantially affect p_{CO_2}, even during intense phytoplankton blooms, as occurred at Murature Beach the following year. Therefore, extreme p_{CO_2} pulse magnitudes may primarily be attributed to spatial differences in geothermal CO_2 degassing across the bay. Fumarole Bay and Murature Beach exhibit stronger outgassing (up to $\sim 7,000\,\mu$atm and $\sim 84,000\,\mu$atm) associated with an underlying magma body and surface thermal anomalies. In contrast, Pendulum Cove and Whalers Bay show lower CO_2 emissions (up to $\sim 2,000\,\mu$atm), which can be primarily associated with residual post-eruptive activity from the 1967–1970 volcanic events (Valentin et al., 1989).

The p_{CO_2} temporal variability cannot be explained by changes in CO_2 solubility alone. Temperature and p_{CO_2} vary almost in-phase at all sites, and not in anti-phase, rather than in anti-phase, which would be expected if temperature changes were the primary driver of p_{CO_2} variability. Temperature and salinity ranges observed in records yields seawater p_{CO_2} variations of $0.2\,\mu$atm. This variation is several orders of magnitude smaller than the measured p_{CO_2} levels, indicating that the direct effects of temperature and salinity on CO_2 solubility are negligible in the observed variability.

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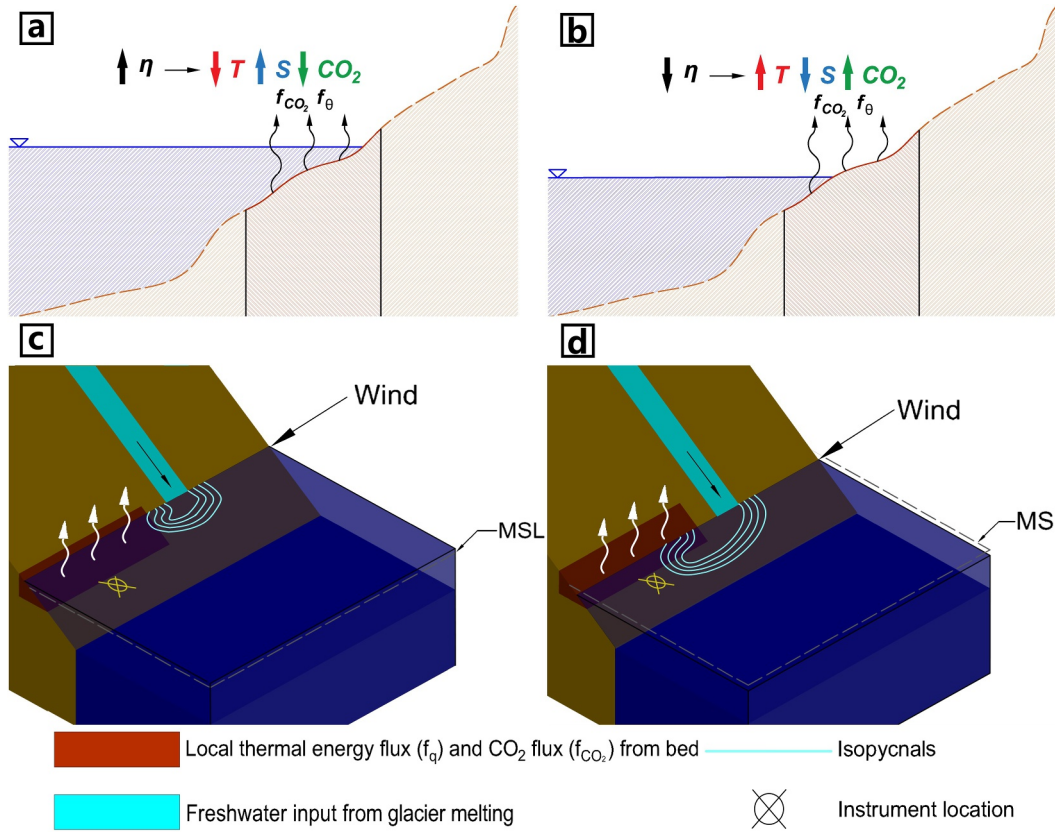


Figure 8. Conceptual sketch of the variability at tidal scales of elevation η, temperature T, salinity S and pCO_2 at Pendulum Cove study site (panels a and c at HW and b and d at LW). In panels (c and d) Mean Sea Level (MSL), defined as the average height of the sea surface over a long period of time, is indicated.

4.1.2. p_{CO_2} Temporal Variability Due To Tides

Seawater p_{CO_2} exceedances relative to site-specific p_{CO_2} background levels (Figure 2) were mainly controlled by tides. Strong coherence and cross-correlations between biogeochemical and physical variables at semidiurnal and diurnal frequencies (Figure 3) indicate that tides are a key control on dissolved CO_2 variability and air–sea exchange. Nearshore fumaroles of Deception Island are a permanent source (at the working time scales in this study) of thermal energy and p_{CO_2} in the water column. In coastal hydrothermal vents, lower water levels (HLW and LLW) correspond to higher temperature and p_{CO_2}, while higher levels (HHW and LHW) reduce concentration and thermal energy (Figure 8, panels a–b). These patterns can be explained by tidal mixing and heat transfer. During flood phases, nearshore waters are mixed and diluted with less concentrated waters. High water levels bring colder and less p_{CO_2} concentrated waters to the coast. For given fluxes of thermal energy and CO_2 from the hydrothermal vent to the overlying water, a water column with a greater depth increases its temperature and p_{CO_2} at a lower rate than one with a shallower depth (Figure 8).

The behavior described above suggests that local changes in CO_2 concentrations (and temperature) are primarily driven by changes in horizontal gradients of CO_2 and thermal energy. The horizontal gradients are maintained by the CO_2 and thermal energy input from the hydrothermal vents, and the contribution within Port Foster is negligible (Caballero-Martínez, Zarzuelo, et al., 2026). Such strong tidal modulation of temperature and p_{CO_2} time series aligns with findings in other coastal systems, such as the Weddell Sea (Droste et al., 2022), where the variability of temperature, salinity, and dissolved inorganic carbon is attributed to advection at semidiurnal periods. In other deeper study areas, tides affect p_{CO_2} in the water column due to the variation of the hydrostatic pressure over the seabed (Li et al., 2019; Zhang et al., 2025). Higher water levels increase hydrostatic pressure on the seabed, thereby reducing gas emissions. Conversely, low water levels decrease hydrostatic pressure, facilitate fracture dilation, allow pore gas pressure to exceed total stress, and trigger rapid gas discharge. Although this

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effect may be present in Port Foster, its contribution would be much smaller than that of CO_2 and thermal energy dissipation in driving p_{CO_2} temporal variability.

4.1.3. p_{CO_2} Temporal Variability Due To Wind, Waves and Glacier Melting

Tides explain a significant fraction of the variability in CO_2 concentration (as well as in temperature, salinity, and DO). Nevertheless, non-tidal processes at event scale also play an important role. At Pendulum Cove, drastic changes in temperature and salinity observed on 02/23/24 (blue rectangle in Figure 4) are most likely driven by wind-induced advection of a meltwater plume. Thermal drone imagery (Figure 5a) provides local spatial context by capturing both the hydrothermal anomaly and the cooler freshwater inland inflow. This freshwater plume produces a nearshore salinity gradient, with lower salinity near the mouth and along the adjacent coast than offshore. Such freshwater influence is negligible at Fumarole and Whalers Bay (Figures 5b and 5c) due to the minor meltwater contribution. Tides modulate the salinity field and strain the water density structure, as schematically shown in Figure 8 (panels c–d). The local increase in salinity at higher water levels (HHW and LHW) results from the intrusion of saltier deeper waters into the nearshore. This pulsating plume behavior is, as discussed above, accompanied by decreases (increases) in temperature and p_{CO_2} as water level increases (decreases). Salinity changes do not appear to result from biogeochemical fluxes from surface soils.

At the event scale, storms and strong winds can drive wind-waves and alter local temperature, salinity, and p_{CO_2} (Llanillo et al., 2019). During the 03/05/24 event at Fumarole Bay (purple rectangle in Figure 6), onshore winds exceeding $10\,m\,s^{-1}$ (panels c–d) generated maximum wave heights up to $H_{max} = 0.9\,m$ (panel e). The storm suppressed tidal variability in the observations, driving drastic changes in temperature (panel a), salinity (panel b), and p_{CO_2} (panel f), enhancing wind-driven advection and promoting dilution through intensified turbulent mixing. Consequently, temperature, salinity, and p_{CO_2} time series converged toward relatively constant values, which were the same as those observed in the main water body of Port Foster, with tidal variability becoming negligible. Despite the short duration of the event, panels a–b and f in Figure 6 indicate a rapid recovery to pre-storm conditions.

4.2. Comparison of Air–Sea CO_2 Fluxes in Port Foster

Air–sea CO_2 fluxes $F(CO_2)$ at Port Foster hydrothermal sites show high-frequency fluctuations, consistent with other study areas (Song et al., 2025). Murature Beach had the highest emission peaks, while Whalers Bay had the lowest peaks, though fluxes there remained high. The air–sea CO_2 fluxes measured in this study consistently show positive values and exceed those reported by Belyaev et al. (2024), who observed negative fluxes in areas farther from the coast. Specifically, these authors reported fluxes in Port Foster ranging between −27 and $52\,mmol\,m^{-2}\,d^{-1}$ during January–February 2021 using the gas transfer formulation by Wanninkhof (2014) under open-sea conditions. However, flux estimates shown in Section 3.2 exhibit a much wider range, extending from ~ -13 to $11,260\,mmol\,m^{-2}\,d^{-1}$. Three factors may explain this discrepancy: (a) Belyaev et al. (2024) collected discrete offshore surface samples are beyond the direct influence of shallow hydrothermal vents; (b) possible inter-annual unaccounted variability between study periods; and (c) methodological differences, notably the use of discrete p_{CO_2} samples without specifying whether sampling occurred during low or high tide.

While extreme p_{CO_2} peaks ($\sim 84,000\,\mu atm$) and associated fluxes ($\sim 11,260\,mmol\,m^{-2}\,d^{-1}$) at Murature Beach exceeded the SAMI-CO_2 validated range ($\sim 2,000\,\mu atm$), pulse timing and relative amplitudes are robust. Even if p_{CO_2} were capped at $\sim 2,000\,\mu atm$, all sites would remain strongly supersaturated, yielding fluxes consistent with active shallow volcanic systems like El Hierro (Fraile-Nuez et al., 2012) and Vulcano (Aiuppa et al., 2021).

Seawater p_{CO_2} values reported in this study (Figure 2) are substantially higher than those reported for other Antarctic coastal areas, such as Marian Cove on King George Island or the Gerlache Strait, where surface p_{CO_2} remains below $500\,\mu atm$ (Mo et al., 2023; Monteiro et al., 2020). In order to put in a wider context the local CO_2 fluxes observed in the waters of Deception Island, a comparison with other polar environments is presented in Table 2. Not only Antarctic but also Arctic and other volcanic-influenced waters are considered. Values shown in Table 2 indicate that the air–sea CO_2 fluxes reported in this study are substantially higher (i.e., outgassing behavior) than other polar environments. Uptake of CO_2 occurs in most of the SO, reaching up to $\sim -250\,mmol\,m^{-2}\,d^{-1}$ in Terra Nova Bay (DeJong & Dunbar, 2017). The lack of hydrothermal influence,

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Table 2
$F(CO_2)$ Fluxes at Different Coastal, Antarctic, Arctic and Volcanic Regions

Location	$F(CO_2)$ $[\text{mmol m}^{-2}\text{ d}^{-1}]$	Polar region	Hydrothermal or geothermal activity
Whalers Bay (Deception Island)[a]	2−228	Antarctica	Yes
Fumarole Bay (Deception Island)[a]	3−529	Antarctica	Yes
Pendulum Cove (Deception Island)[a]	4−134	Antarctica	Yes
Murature Beach (Deception Island)[a]	−13−11,257[b]	Antarctica	Yes
Port Foster (Deception Island)[c]	−27−52	Antarctica	Yes
Marian Cove (King George Island)[d]	−23−30	Antarctica	No
Terra Nova Bay (Ross Sea)[e]	−246−30	Antarctica	No
Gerlache Strait (Palmer Archipelago)[f]	−23−3	Antarctica	No
Ryder Bay (Western Antarctica)[g]	∼−60− ∼ 0	Antarctica	No
Marguerite Bay (Western Antarctica)[h]	−45−(−42)	Antarctica	No
Bellingshausen Sea (Western Antarctica)[i]	−9−6	Antarctica	No
Amundsen Sea (Western Antarctica)[j]	−52−25	Antarctica	No
Mertz Polynya (Eastern Antarctica)[k]	∼ −15	Antarctica	No
Enderby Basin (Eastern Antarctica)[l]	−81−(−2)	Antarctica	No
Prydz Bay Basin (Eastern Antarctica)[m]	−49−6	Antarctica	No
Beaufort Sea (Western Arctic)[n]	−18−3	Arctic	No
East Siberian Sea (Eurasian Arctic)[o]	−15−9	Arctic	No
Guanabara Bay (Brazil)[p]	26−49	No	No
Mar Menor (Spain)[q]	2−5	No	No
Bay of Cádiz inner sack (Spain)[r]	−8−31	No	No
Mount Peteroa (Argentina)[s]	140−2,600	No	Yes
Vulcano Island (Italy)[t]	up to 161,000	No	Yes

Note. Antarctic and Arctic fluxes are refereed to summer conditions. Values are rounded to whole units to improve readability, facilitate comparability, and evaluate consistency across the results of different studies. [a]From this work. Coordinates of each location are shown in Table 1. [b]Peak values come from p_{CO_2} observations above the validated range of the sensor and are order-of-magnitude estimates (see Section 2.2). [c]Belyaev et al. (2024). Coordinates: 62°58′00″ S-60°39′00″ W. [d]Mo et al. (2023). Coordinates: 62°13′00″ S-58°48′00″ W. [e]DeJong and Dunbar (2017). Coordinates: 74°50′00″ S-164°30′00″ E. [f]Monteiro et al. (2020). Coordinates: 64°30′00″ S-62°20′00″ W. [g]Droste et al. (2025). Coordinates: 67°34′50″ S-68°10′50″ W. [h]Jones et al. (2017). Coordinates: 68°30′00″ S-68°30′00″ W. [i]Ruiz-Halpern et al. (2014). Coordinates: 71°0′00″ S-85°00′00″ W. [j]Mu et al. (2014). Coordinates: 73°00′00″ S-112°00′00″ W. [k]Shadwick et al. (2014). Coordinates: 67°30′00″ S-145°00′00″ E. [l]Shetye et al. (2017). Coordinates: 66−70°00′00″ S-20−70°00′00″ E. [m]Wang et al. (2021). Coordinates: 67°00′00″ S-75°00′00″ E. [n]Murata and Takizawa (2003). Coordinates: 68−75°00′00″ N-127−170°00′00″ W. [o]Pipko et al. (2017). Coordinates: 70−85°00′00″ N-8−167°00′00″ E. [p]Cotovicz Jr. et al. (2015). Coordinates: 22°47′26″ S-43°09′20″ W. [q]Vallejo et al. (2021). Coordinates: 37°45′00″ N-0°47′00″ W. [r]Amaya-Vías et al. (2025). Coordinates: 36°29′39″ N-6°13′01″ W. [s]Sanci et al. (2010). Coordinates: 35°14′57″ S-70°30′23″ W. [t]Di Martino et al. (2022) and Inguaggiato et al. (2012). Coordinates: 38°24′15″ N-14°57′57″ E.

together with sea-ice cover during long periods in polar regions, reduces outgassing and strengthens their role as CO_2 sinks (Brown et al., 2019). Negative observed fluxes (CO_2 uptake) are common across Arctic regions, although these tend to be larger in magnitude in Antarctic waters (Murata & Takizawa, 2003; Pipko et al., 2017). Furthermore, Port Foster fluxes exceed those reported for other coastal regions worldwide, including Bay of Cádiz and Mar Menor (Spain), and Guanabara Bay (Brazil) (Amaya-Vías et al., 2025; Cotovicz Jr. et al., 2015; Vallejo et al., 2021). Observations in hydrothermal sites in Port Foster are on the same order of magnitude measurements in volcanic sites, such as Mount Peteroa ($\sim 2,600\ \text{mmol m}^{-2}\text{ d}^{-1}$; Sanci et al. (2010)), and lower than those reported for highly active volcanic systems such as Vulcano Island (Italy), where fluxes attain values from 32,200 and to $161,600\ \text{mmol m}^{-2}\text{ d}^{-1}$ (Di Martino et al., 2022; Inguaggiato et al., 2012). The fluxes

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obtained in Deception Island are closer to those at other shallow water hydrothermal systems and gas emission sites worldwide than to those of other polar, non volcanic sites.

The nearshore hydrothermal sites monitored at Deception Island exhibited mostly positive air–sea CO_2 fluxes, even though the global ocean is typically a net CO_2 sink (Fay et al., 2024). This local outgassing behavior reflects their direct exposure to volcanic CO_2 inputs and should not be extrapolated to the basin-wide CO_2 balance of Port Foster. Similar patterns have been observed in other submarine volcanoes, where $F(CO_2)$ of $6.0 \cdot 10^5 \pm 1.1 \cdot 10^5$ kg d^{-1} ($\sim 1.05 \cdot 10^6 \pm 0.2 \cdot 10^6$ mmol m^{-2} d^{-1} in a surface of 13,000 $\pm$ 800 m^2) were estimated in seawater (Santana-Casiano et al., 2016) or diffuse CO_2 emissions of up to 190.9 mmol m^{-2} d^{-1} were measured in volcanic lakes (Andrade et al., 2025).

5. Conclusions

This study improves the understanding of the spatio-temporal variability of seawater p_{CO_2} and air–sea CO_2 fluxes at four coastal hydrothermal sites in Port Foster (Deception Island, Antarctica), namely Pendulum Cove, Fumarole Bay, Whalers Bay, and Murature Beach.

The four locations showed broadly similar physical controls but differed in the relative influence of biological processes on p_{CO_2}, possibly reflecting contrasting phytoplankton bloom conditions in 2024 and 2025 sampling periods. The p_{CO_2} temporal variability is mainly driven by tidal and wind-wave mixing, wind-induced advection, and meltwater runoff. Background levels were strikingly similar at all locations. Pendulum Cove, Fumarole Bay, and Whalers Bay exhibited values ranging between 600 and 640 μatm, all of them above atmospheric concentration. The low Chl-*a* concentrations observed at these sites in the 2024 survey indicated the absence of a phytoplankton bloom and a limited influence of primary production on minimum p_{CO_2} values. In contrast, the high Chl-*a* concentration measured at Murature Beach suggested the presence of a phytoplankton bloom in February 2025. The associated biological CO_2 uptake likely contributed to the lower background p_{CO_2} value observed at Murature beach (300 μatm), which was below atmospheric levels. Significant pulses of p_{CO_2} were also observed, exceeding background values by several orders of magnitude. Maximum p_{CO_2} values reached up to $\sim$1,500 μatm at Pendulum Cove, $\sim$2,000 μatm at Fumarole Bay, and $\sim$7,000 μatm at Whalers Bay. Murature Beach showed the highest nominal peak values of the four sites, reaching up to 84,000 μatm.

At all four hydrothermal sites, high p_{CO_2} values occurred during low tides, while low values occurred during high tides when nearshore waters mixed with those less concentrated from Port Foster. Therefore, local CO_2 variability is mainly controlled by tidally-driven horizontal mixing acting on the persistent CO_2 gradient maintained by the hydrothermal source. Additionally, meltwater reduces locally salinity and temperature, affecting vertical air–sea fluxes and creating horizontal density gradients. Meltwater plume size is shaped by tidal mixing and wind-driven advection, which is able to transport biogeochemical properties near the surface. Wind also destabilizes the water column, enhancing p_{CO_2} dilution, especially during storms.

Tides, wind, wind-waves, and meltwater inputs control air–sea fluxes at all locations. Air–sea fluxes estimated at all locations exhibit tidally-induced pulses above their respective background values, similarly to p_{CO_2} observations. Air–sea flux pulses reach peaks on the order of magnitude of $\sim$130 mmol m^{-2} d^{-1}, $\sim$530 mmol m^{-2} d^{-1}, and $\sim$230 mmol m^{-2} d^{-1} at Pendulum Cove, Fumarole Bay, and Whalers Bay, respectively. Remarkably, Murature Beach exhibited both negative fluxes down to $\sim$–13 mmol m^{-2} d^{-1} and extreme positive peaks reaching up to 11,260 mmol m^{-2} d^{-1}, which are one to two orders of magnitude higher than those at other sites. Compared to other polar and coastal areas, air–sea CO_2 fluxes at Deception Island are substantially higher, and might be similar in magnitude to fluxes measured in other active volcanic systems worldwide.

Overall, this study reveals exceptionally high p_{CO_2} concentrations and associated fluxes. These findings underscore the importance of local physical forcings in controlling air–sea CO_2 exchange in a crucial, under-observed volcanic region of the Southern Ocean, considered one of the most active regions on the Antarctic continent. This insight into unique ecosystems is vital to quantify CO_2 stocks, which is essential to predict future trends and help implement mitigation strategies for global warming.

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The authors agree that all observational data compiled for this study, along with the processed and derived data sets supporting the results and conclusions, are available in an open-access repository through this link: http://doi.org/10.6084/m9.figshare.31123825.

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